Verified formulas of the General Sinus, General Numbers and Cosmic Mass theory

Work preparation package for the virtual universe

10 November 2026

Every formula below was checked by a script of `02-verification-scripts` or by the 50-digit audit of this package. The tag in square brackets names the source:

- **GS:** General Sinus, revision 3, and the book's Appendix A;
- **GN:** the General Numbers paper and the book's Chapter 4;
- **CM:** the Cosmic Mass paper and the book's Chapters 6 and 9.

Formulas found false are listed only in Section 4, with their corrections.

1 General Sinus: trigonometry of an oblique projection

Definitions. The point Z on the second side of the angle α is projected onto the first side along a line at the angle ϕ to ZO:

$$\sin(\alpha,\phi)=\frac{\sin\alpha}{\sin(\alpha+\phi)},\quad \cos(\alpha,\phi)=\frac{\sin\phi}{\sin(\alpha+\phi)},\quad \tan(\alpha,\phi)=\frac{\sin\alpha}{\sin\phi},$$

$$\csc(\alpha,\phi)=\frac{\sin(\alpha+\phi)}{\sin\alpha},\quad \sec(\alpha,\phi)=\frac{\sin(\alpha+\phi)}{\sin\phi},\quad \cot(\alpha,\phi)=\frac{\sin\phi}{\sin\alpha}.[GS]$$

For $\alpha+\phi=\pi/2$ the classical functions return.

Coordinate reading and Pythagoras. With $\overline{\phi}=\alpha+\phi$ and $\vec{u}=(\cos\overline{\phi},\sin\overline{\phi})$,

$$(\cos\alpha,\sin\alpha)=\cos(\alpha,\phi)\,\vec{e}_1+\sin(\alpha,\phi)\,\vec{u},$$
$$\cos(\alpha,\phi)^2+\sin(\alpha,\phi)^2+2\cos(\alpha,\phi)\sin(\alpha,\phi)\cos(\alpha+\phi)=1.$$

[GS]

Exchange, reciprocals, signs.

$$\sin(\alpha,\phi)=\cos(\phi,\alpha),\quad \tan(\alpha,\phi)=\cot(\phi,\alpha),\quad \csc(\alpha,\phi)=\sec(\phi,\alpha);$$

$$\csc(\alpha,\phi)=\sin(\alpha+\phi,-\phi),\quad \sec(\alpha,\phi)=\tan(\alpha+\phi,\phi),\quad \cot(\alpha,\phi)=\cos(\alpha-\phi,\phi);$$

$$\sin(-\alpha,-\phi)=\sin(\alpha,\phi),\quad \cos(-\alpha,-\phi)=\cos(\alpha,\phi),\quad \sin(-\alpha,\phi)=\sin(\alpha,-\phi),$$
$$\tan(i\alpha,j\phi)=ij\tan(\alpha,\phi)\quad (i,j=\pm1);$$
$$\sin(\alpha+p\pi,\phi+q\pi)=(-1)^q\sin(\alpha,\phi),\quad \cos(\alpha+p\pi,\phi+q\pi)=(-1)^p\cos(\alpha,\phi),$$
$$\tan(\alpha+p\pi,\phi+q\pi)=(-1)^{p+q}\tan(\alpha,\phi).$$

[GS]

Special values. $\sin(\alpha, \frac{\pi}{2}) = \tan\alpha$, $\cos(\alpha, \frac{\pi}{2}) = \sec\alpha$, $\tan(\alpha, \frac{\pi}{2}) = \sin\alpha$; $\sin(\theta,\theta) = \cos(\theta,\theta) = \frac{1}{2\cos\theta}$, $\tan(\theta,\theta) = 1$; all six equal 1 at $\theta = \pi/3$; $\sin(\frac{\pi}{5}, \frac{\pi}{5}) = \tau^{-1}$ with $\tau = (1+\sqrt{5})/2$. [GS]

Ptolemy's identity. $\sin(x+y)\sin(y+z) = \sin x \sin z + \sin y \sin(x+y+z)$. Every addition formula below reduces to it. [GS]

Addition with the projection angle fixed (moving frame).

$$\begin{aligned}
\sin(\alpha+\beta,\phi) &= \sin(\alpha,\phi) + \cos(\alpha,\phi)\sin(\beta,\phi+\alpha)\\
&= \sin(\alpha,\phi) + \cos(\alpha,\phi)\sin(\beta,\phi)\cos(\alpha,\phi+\beta),\\
\cos(\alpha+\beta,\phi) &= \cos(\alpha,\phi)\cos(\beta,\phi) + \sin(\alpha,\phi)\sin(\beta,\phi)\cos(\alpha+\beta,\phi)\\
&= \frac{\cos(\alpha,\phi)\cos(\beta,\phi)}{1-\sin(\alpha,\phi)\sin(\beta,\phi)}\\
&= \cos(\alpha,\phi)\cos(\beta,\phi) + \cos(\alpha,\phi)\sin(\beta,\phi)\sin(\alpha,\phi+\beta),\\
\sec(\alpha+\beta,\phi) &= \sec(\alpha,\phi)\sec(\beta,\phi) - \tan(\alpha,\phi)\tan(\beta,\phi),\\
\tan(\alpha+\beta,\phi) &= \tan(\alpha,\phi)\cos\beta + \tan(\beta,\phi)\cos\alpha.
\end{aligned}$$

[GS]

Addition in the projection angle. $\sin(\alpha,\phi+\theta) = \sin(\alpha,\phi)\sin(\alpha+\phi,\theta)$, $\cos(\alpha,\phi+\theta) = \cos(\alpha,\phi) + \sin(\alpha,\phi)\cos(\alpha+\phi,\theta)$. [GS]

Double and half angles.

$$\begin{gathered}
\sin(2\alpha,\phi) = \sin(\alpha,\phi)\big(1+\cos(2\alpha,\phi)\big), \quad \cos(2\alpha,\phi) = \frac{\cos(\alpha,\phi)^2}{1-\sin(\alpha,\phi)^2},\\
\sec(2\alpha,\phi) = \sec(\alpha,\phi)^2 - \tan(\alpha,\phi)^2,\\
\tan(2\alpha,\phi) = 2\tan(\alpha,\phi)\cos\alpha, \quad \sin(2\alpha,\phi) - \sin(\alpha,\phi) = \sin(\alpha,\phi)\cos(2\alpha,\phi),\\
\cos(\alpha,\phi) = \frac{\cos(\frac{\alpha}{2},\phi)^2}{1-\sin(\frac{\alpha}{2},\phi)^2}, \quad \sec(\alpha,\phi) = \sec(\tfrac{\alpha}{2},\phi)^2 - \tan(\tfrac{\alpha}{2},\phi)^2.
\end{gathered}$$

[GS]

Fixed frame (Euler and De Moivre). For fixed $\overline{\phi} \in (0,\pi)$, write $C_\theta = \sin(\overline{\phi}-\theta)/\sin\overline{\phi}$, $S_\theta = \sin\theta/\sin\overline{\phi}$ and $u = e^{i\overline{\phi}}$. Then

$$\begin{gathered}
e^{i\theta} = C_\theta + S_\theta u, \qquad u^2 = 2\cos\overline{\phi}\, u - 1,\\
C_{\alpha+\beta} = C_\alpha C_\beta - S_\alpha S_\beta, \qquad S_{\alpha+\beta} = C_\alpha S_\beta + S_\alpha C_\beta + 2\cos\overline{\phi}\, S_\alpha S_\beta,\\
(C_\theta + S_\theta u)^n = C_{n\theta} + S_{n\theta} u,\\
u^n = -U_{n-2}(\cos\overline{\phi}) + U_{n-1}(\cos\overline{\phi})\, u \quad (n \geq 2).
\end{gathered}$$

[GS]

Partial derivatives (corrected second order).

$$\partial_x \sin(x,y) = \frac{\cos(x,y)}{\sin(x+y)}, \quad \partial_y \sin(x,y) = -\sin(x,y)\cot(x+y),$$
$$\partial_x \cos(x,y) = -\cos(x,y)\cot(x+y), \quad \partial_y \cos(x,y) = \frac{\sin(x,y)}{\sin(x+y)},$$
$$\partial_x \tan(x,y) = \frac{\cos x}{\sin y} = \tan(\tfrac{\pi}{2} \pm x, y), \quad \partial_y \tan(x,y) = -\tan(x,y)\cot y,$$
$$\partial_x^2 \sin = -2\cot(x+y)\,\partial_x \sin, \quad \partial_y^2 \sin = \sin(x,y)\big(1+2\cot^2(x+y)\big),$$
$$\partial_x^2 \cos = \cos(x,y)\big(1+2\cot^2(x+y)\big), \quad \partial_y^2 \cos = -2\cot(x+y)\,\partial_y \cos,$$
$$\partial_x^2 \tan = -\tan(x,y), \quad \partial_y^2 \tan = \tan(x,y)\big(1+2\cot^2 y\big).$$

[GS]

Derivative with the frame fixed. $Df = \partial_x f - \partial_y f$ (path $x+t$, $y-t$). For every integer $n \geq 0$:

$$D^n \sin(x,y) = \sin\big(x + \tfrac{n\pi}{2},\, y - \tfrac{n\pi}{2}\big), \qquad D^n \cos(x,y) = \cos\big(x + \tfrac{n\pi}{2},\, y - \tfrac{n\pi}{2}\big), \qquad D^2 f = -f,$$
$$D^n \sin\big(x - \tfrac{n\pi}{2},\, y + \tfrac{n\pi}{2}\big) = \sin(x,y) \text{ (primitives)},$$
$$D^n \tan(x,y) = \sin(x+y)\Big(-\frac{d}{dy}\Big)^{n-1} \csc^2 y \quad (n \geq 1).$$

[GS]

Products for every $N \geq 1$.

$$\sin(x,y) = \prod_{k=0}^{N-1} \sin\Big(\tfrac{x+\pi k}{N}, \tfrac{y}{N}\Big), \quad \cos(x,y) = \prod_{k=0}^{N-1} \cos\Big(\tfrac{x}{N}, \tfrac{y+\pi k}{N}\Big),$$
$$\tan(x,y) = \prod_{k=0}^{N-1} \tan\Big(\tfrac{x+\pi k}{N}, \tfrac{y+\pi k}{N}\Big).$$

[GS]

General Sinus Transform. Take $P_{N,k}(z) = u_k z^2 - 2N v_k z - 2N^2 u_k$ with exact $u_k = \sin(\pi k/N)$ and $v_k = \cos(\pi k/N)$, and set $S_N = \prod_k P_{N,k}(x)/P_{N,k}(x+y)$ (C_N and T_N likewise). With $\rho(\varepsilon) = \sqrt{1+\varepsilon^4/4}$ and $\delta(\varepsilon) = \arctan \frac{\varepsilon}{1-\varepsilon^2/2}$, for $N > \max(|x|,|y|,|x+y|)/\sqrt{2}$:

$$S_N(x,y) = \Big(\frac{\rho(x/N)}{\rho((x+y)/N)}\Big)^N \frac{\sin(N\delta(x/N))}{\sin(N\delta((x+y)/N))},$$
$$S_N = \sin(x,y) + \frac{x^3 \cos x \sin(x+y) - (x+y)^3 \sin x \cos(x+y)}{6N^2 \sin^2(x+y)} + O(N^{-3}).$$

[GS]

2 General Numbers: the plane in a leaning frame

Axiom and the two angles of one length. $\mathcal{G} = |\mathcal{G}|\,\vec{u}$, $\vec{u} = (\cos\overline{\phi}, \sin\overline{\phi})$, $|\mathcal{G}| = |\cos\overline{\phi} + \sin\overline{\phi}| = \sqrt{2}\,|\sin(\overline{\phi} + \frac{\pi}{4})|$. For $1 < A < \sqrt{2}$: $\overline{\phi}_1 = \frac{1}{2}\arcsin(A^2-1)$, $\overline{\phi}_2 = \frac{\pi}{2} - \overline{\phi}_1$ ($A = 1.16043$: $10.14°$ and $79.86°$). [GN]

Rule of the Square and the product. For $z = a + b\mathcal{G}$ and $w = c + d\mathcal{G}$:

$$\mathcal{G}^2 = 2|\mathcal{G}|\cos\overline{\phi}\,\mathcal{G} - |\mathcal{G}|^2, \qquad zw = (ac - |\mathcal{G}|^2 bd) + (ad + bc + 2|\mathcal{G}|\cos\overline{\phi}\,bd)\,\mathcal{G}. [GN]$$

Readings. $\operatorname{Re} z = a$, $\operatorname{Com} z = b\mathcal{G}$, $\operatorname{Sh} z = a + b|\mathcal{G}|\cos\overline{\phi}$, $\operatorname{Ht} z = b|\mathcal{G}|\sin\overline{\phi}$, and $|z|^2 = a^2 + 2ab|\mathcal{G}|\cos\overline{\phi} + b^2|\mathcal{G}|^2 = (\operatorname{Sh} z)^2 + (\operatorname{Ht} z)^2$. [GN]

Conjugate, skew flip, division. $\bar{z} = (a+2b|\mathcal{G}|\cos\overline{\phi}) - b\mathcal{G}$, $z + \bar{z} = 2\operatorname{Sh} z$, $\operatorname{Re}\bar{z} = 2\operatorname{Sh} z - \operatorname{Re} z$, $z\bar{z} = |z|^2$, $1/z = \bar{z}/|z|^2$. The skew flip is $a - b\mathcal{G}$, and $(a+b\mathcal{G})(a-b\mathcal{G}) = a^2 + b^2|\mathcal{G}|^2 - 2b^2|\mathcal{G}|\cos\overline{\phi}\,\mathcal{G}$. The norm is multiplicative. [GN]

Turning Unit and the bridge to $\mathbb{C}$. $\mathcal{J} = -\cot\overline{\phi} + \mathcal{G}/(|\mathcal{G}|\sin\overline{\phi})$, $\mathcal{J}^2 = -1$, $z = \operatorname{Sh} z + \operatorname{Ht} z\,\mathcal{J}$; the map $z \mapsto \operatorname{Sh} z + i\operatorname{Ht} z$ is an isomorphism of fields onto $\mathbb{C}$. [GN]

Laws of the parts. $\operatorname{Re}(zw) = \operatorname{Re} z \operatorname{Re} w - |\mathcal{G}|^2 bd$, $\operatorname{Com}(zw) = \operatorname{Sh}(z)\operatorname{Com}(w) + \operatorname{Sh}(w)\operatorname{Com}(z)$, $\operatorname{Sh}(zw) = \operatorname{Sh} z \operatorname{Sh} w - \operatorname{Ht} z \operatorname{Ht} w$, $\operatorname{Ht}(zw) = \operatorname{Sh} z \operatorname{Ht} w + \operatorname{Sh} w \operatorname{Ht} z$, $|\operatorname{Re} z| \le |z|/\sin\overline{\phi}$ (the bound is reached). [GN]

Turning, powers, roots. $e^{\theta\mathcal{J}} = C_\theta + S_\theta\vec{u}$; $(C_\theta + S_\theta\vec{u})^n = C_{n\theta} + S_{n\theta}\vec{u}$; $\mathcal{G}^n = -|\mathcal{G}|^n \frac{\sin(n-1)\overline{\phi}}{\sin\overline{\phi}} + |\mathcal{G}|^{n-1}\frac{\sin n\overline{\phi}}{\sin\overline{\phi}}\mathcal{G}$; the powers lie on the spiral $\rho = |\mathcal{G}|^{\theta/\overline{\phi}}$; the n-th roots are $|z|^{1/n}(C_{\theta_k} + S_{\theta_k}\vec{u})$ with $\theta_k = (\alpha + 2\pi k)/n$. [GN]

Cauchy–Riemann in a leaning frame. For $f = U + V\mathcal{G}$ holomorphic: $\partial_b U = -|\mathcal{G}|^2\partial_a V$ and $\partial_b V = \partial_a U + 2|\mathcal{G}|\cos\overline{\phi}\,\partial_a V$. [GN]

Generalized complex numbers. $\mathcal{G}^2 = p + q\mathcal{G}$ with $p = -|\mathcal{G}|^2$ and $q = 2|\mathcal{G}|\cos\overline{\phi}$, so $q^2 + 4p < 0$ (the elliptic case). Conversely, every elliptic pair (p, q) is a leaning frame with length $\sqrt{-p}$ and angle $\arccos(q/2\sqrt{-p})$. [GN]

3 Cosmic Mass

The cosmic mass and its unit. $\mu = m + q\,\mathcal{G}$, with $|\mathcal{G}| = s = \sqrt{K/G} = 1.16043 \times 10^{10}$ kg/C. The drawing length is $A = s/u$, and $|\cos\overline{\phi} + \sin\overline{\phi}| = A$; with $u = 10^{10}$ kg/C, $A = 1.16043$ and $\overline{\phi} = 79.86°$ (or $10.14°$). In Planck or Stoney units $A = 1$ and $\overline{\phi} = 90°$. [CM]

Cosmic Interaction Law and its reading.

$$\vec{F}_G = \frac{G\,\mu_a\mu_b}{r^2}\,\vec{e}_r, \qquad F = \frac{G\,\operatorname{Re}(\mu_a\mu_b)}{r^2} = \frac{Gm_am_b - Kq_aq_b}{r^2} \quad \text{at every angle,}$$
$$\operatorname{Re}\big[(a+b\mathcal{G})(c+d\mathcal{G})\big] = ac - s^2bd.$$

[CM]

Full product (both coordinates).

$$\mu_a\mu_b = (m_am_b - s^2q_aq_b) + \big(m_aq_b + q_am_b + 2s\cos\overline{\phi}\,q_aq_b\big)\mathcal{G}.$$

The Composer coordinate holds the Coupling term $m_aq_b + q_am_b$ and the charge share $2s\cos\overline{\phi}\,q_aq_b$. It is not read as a force: see the book's Chapter 10 and Section 4. [CM]

Stoney mass. $m_S = e\,s \approx 1.85921 \times 10^{-9}$ kg $= \sqrt{\alpha_{\mathrm{fs}}}\,m_P$. [CM]

Cosmic Field Law and law of motion. Fields with General Number values, $\boldsymbol{\mathcal{E}} = \mathbf{g} + \mathbf{e}\,\mathcal{G}$ and $\boldsymbol{\mathcal{B}} = \mathbf{B}_g + \mathbf{b}\,\mathcal{G}$; sources $\rho_\mu = \rho_m + \rho_q\mathcal{G}$ and $\mathbf{J}_\mu = \mathbf{J}_m + \mathbf{J}_q\mathcal{G}$:

$$\nabla\cdot\boldsymbol{\mathcal{E}} = -4\pi G\rho_\mu, \quad \nabla\cdot\boldsymbol{\mathcal{B}} = 0, \quad \nabla\times\boldsymbol{\mathcal{E}} = -\partial_t\boldsymbol{\mathcal{B}}, \quad \nabla\times\boldsymbol{\mathcal{B}} = -\frac{4\pi G}{c^2}\mathbf{J}_\mu + \frac{1}{c^2}\partial_t\boldsymbol{\mathcal{E}},$$

$$\mathrm{Re}(\mu)\frac{d\mathbf{v}}{dt} = \mathrm{Re}\big[\mu(\boldsymbol{\mathcal{E}} + \mathbf{v}\times\boldsymbol{\mathcal{B}})\big] \iff m\frac{d\mathbf{v}}{dt} = m(\mathbf{g} + \mathbf{v}\times\mathbf{B}_g) + q(\mathbf{E} + \mathbf{v}\times\mathbf{B}).$$

Each field equation splits into its two coordinates. The Composer set is Maxwell's, with $\mathbf{E} = -s^2\mathbf{e}$, $\mathbf{B} = -s^2\mathbf{b}$ and $\mu_0 = 4\pi K/c^2$. The field of a source at rest is $\boldsymbol{\mathcal{E}} = -G\mu_s\vec{e}_r/r^2$. [CM]

Kept quantities. $\partial_t\rho_\mu + \nabla\cdot\mathbf{J}_\mu = 0$. Both fields obey the wave equation with speed c.

$$u = -\frac{g^2 + c^2B_g^2}{8\pi G} + \frac{E^2 + c^2B^2}{8\pi K}, \qquad \mathbf{S} = -\frac{c^2}{4\pi G}\,\mathrm{Re}(\boldsymbol{\mathcal{E}}\times\boldsymbol{\mathcal{B}}), \qquad \partial_t u + \nabla\cdot\mathbf{S} = -\,\mathrm{Re}(\boldsymbol{\mathcal{E}}\cdot\mathbf{J}_\mu).$$

The momentum balance holds with the stress T_{ij} of the paper. The action is

$$S = \int dt\,\Big\{\sum\big[\tfrac{1}{2}\,\mathrm{Re}(\mu)v^2 - \mathrm{Re}\big(\mu(\Phi - \mathbf{v}\cdot\boldsymbol{\mathcal{A}})\big)\big] - \frac{1}{8\pi G}\int \mathrm{Re}\big[\boldsymbol{\mathcal{E}}\cdot\boldsymbol{\mathcal{E}} - c^2\boldsymbol{\mathcal{B}}\cdot\boldsymbol{\mathcal{B}}\big]\,dV\Big\}.$$

[CM]

Where the law fails (stated in the paper).

- Frame dragging on LAGEOS comes out at 1/4 of the measured value.
- There is no light bending and no Shapiro delay.
- Gravity waves carry negative energy.
- There is no conversion of rest mass into energy.
- The law is valid for slow bodies only.

[CM]

4 Corrections and exclusions

statement	status	correct form / note
$\tan(2\alpha,\phi)-\sec(2\alpha,\phi)=\sin(\alpha,\phi)$	false	deleted
$\tan(\alpha,\phi)=\sin(\frac{\alpha}{2},\phi)+\sec(\alpha,\phi)$	false	deleted
$\partial_y^2\sin=\sin\left(2\cot^2(x+y)-1\right)$	false	$\sin\left(1+2\cot^2(x+y)\right)$
$\partial_x^2\cos=\cos\left(2\cot^2(x+y)-1\right)$	false	$\cos\left(1+2\cot^2(x+y)\right)$
$\partial_y^2\tan=\tan\left(2\cot^2(x+y)-1\right)$	false	$\tan\left(1+2\cot^2 y\right)$
$D^2\sin=+\sin$, $D^2\cos=+\cos$	false	$D^2 f=-f$
D^n and primitives by $\omega(z,n)$	false for $n\equiv 2,3$	$D^n f(x,y)=f(x+\frac{n\pi}{2},y-\frac{n\pi}{2})$
$\tan^{(n)}=n!\,\tan'\tan(\frac{\pi}{2}+y,y)^{n-1}$	true only $n\le 2$	$D^n\tan=\sin(x+y)(-\frac{d}{dy})^{n-1}\csc^2 y$
rotation recurrence for $u_{n,k}$, $v_{n,k}$ "exact"	false	use exact sines; error accumulates otherwise
five digits for $n\ge 20$ at every point	false with the recurrence	rate $c(x,y)/N^2$, proved (Section 1)
heat-equation analogy, $R^2=0.83$	withdrawn	not evidence
Composer coordinate as a force along the line, full size	excluded by measurement	neutron would carry $9.0\times10^{-19}e$; measured $(-0.2\pm0.8)\times10^{-21}e$; share below ≈ 0.002 (book, Ch. 10)
charge share as a force along the line	excluded	hydrogen would bind at 5.71 eV, measured 13.598 eV (book, Ch. 10)
Standard Model (book, Ch. 7), Closure Law (Ch. 8)	not verified	kept out of the papers; Closure Law is a refitted empirical law